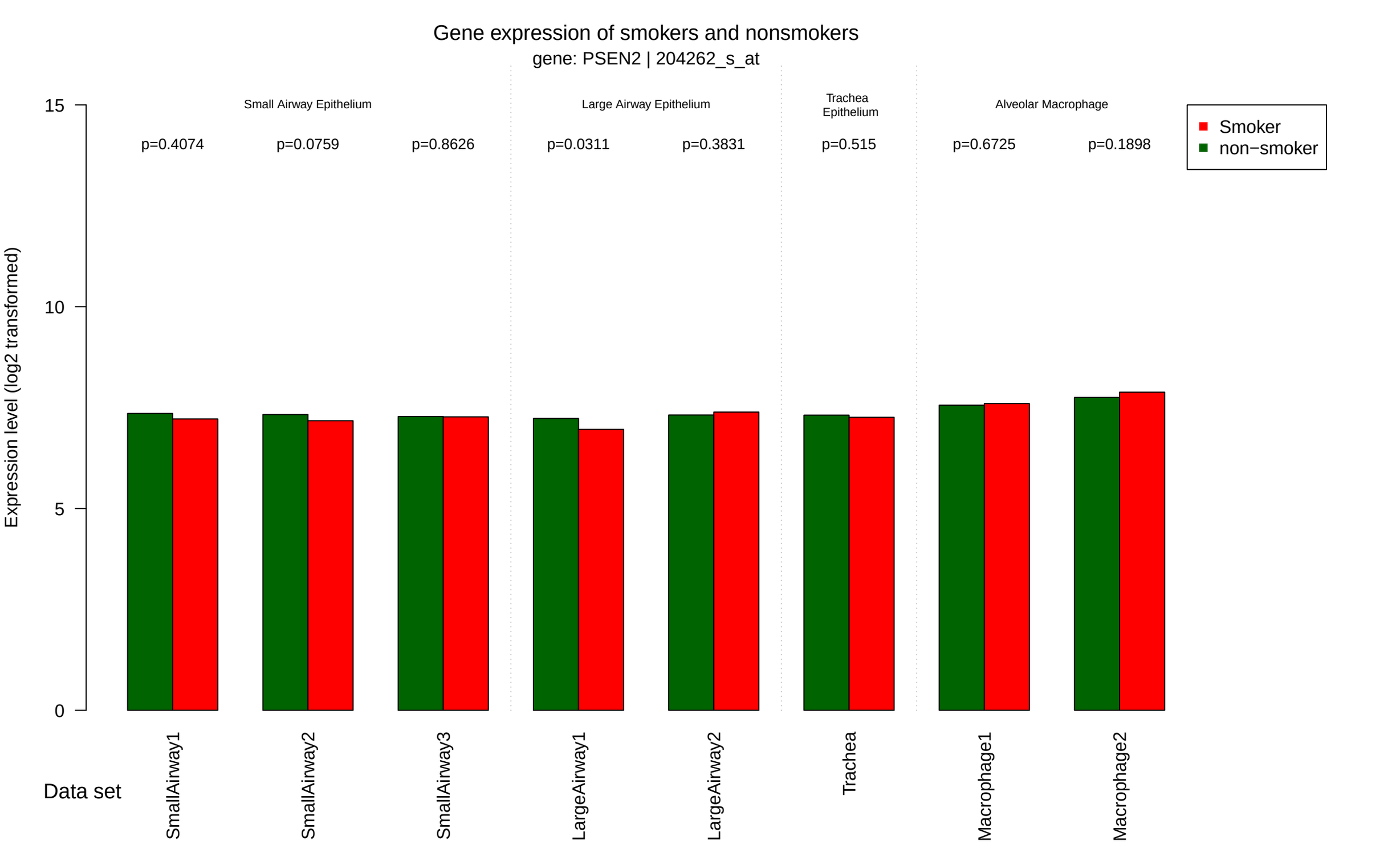
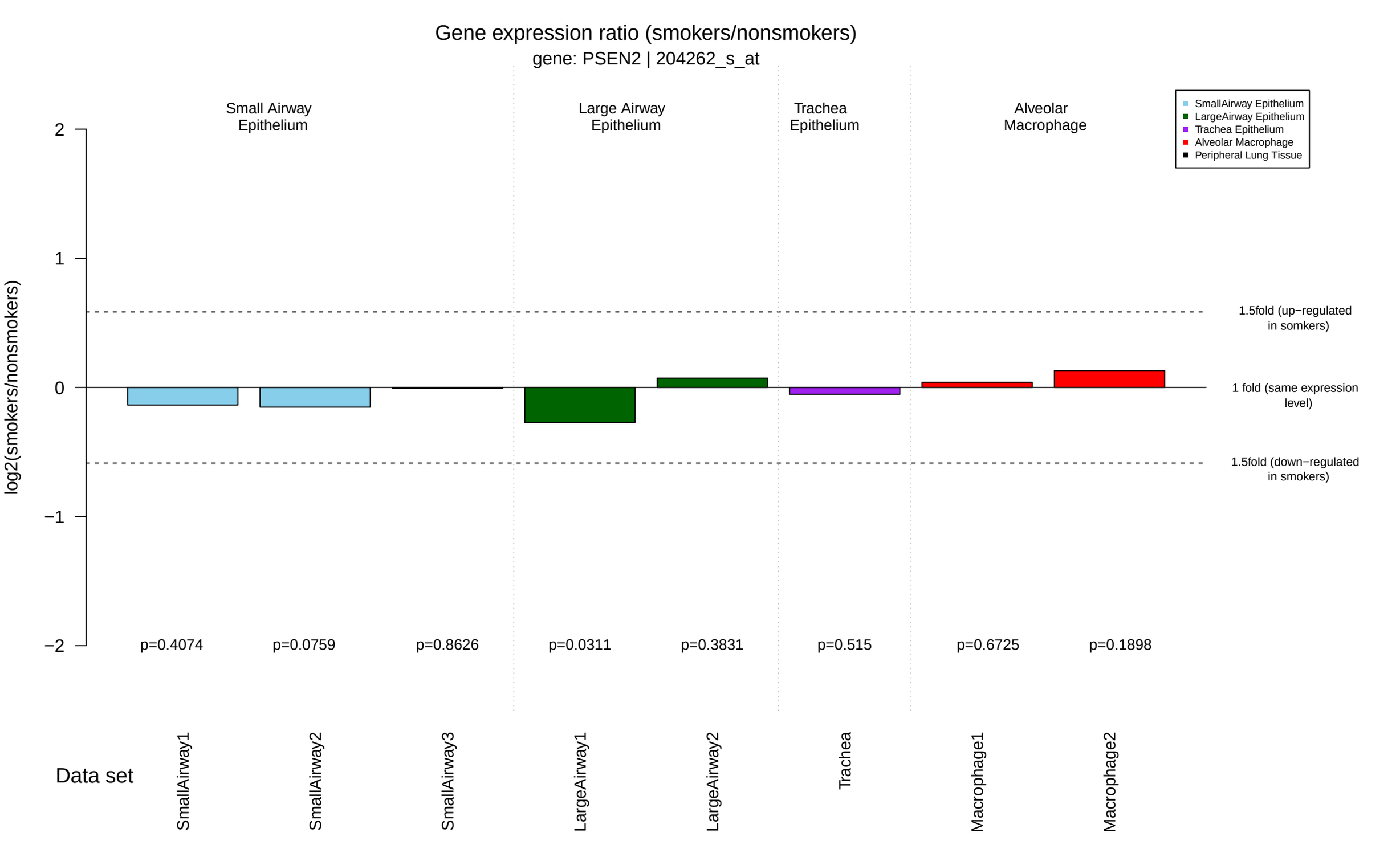
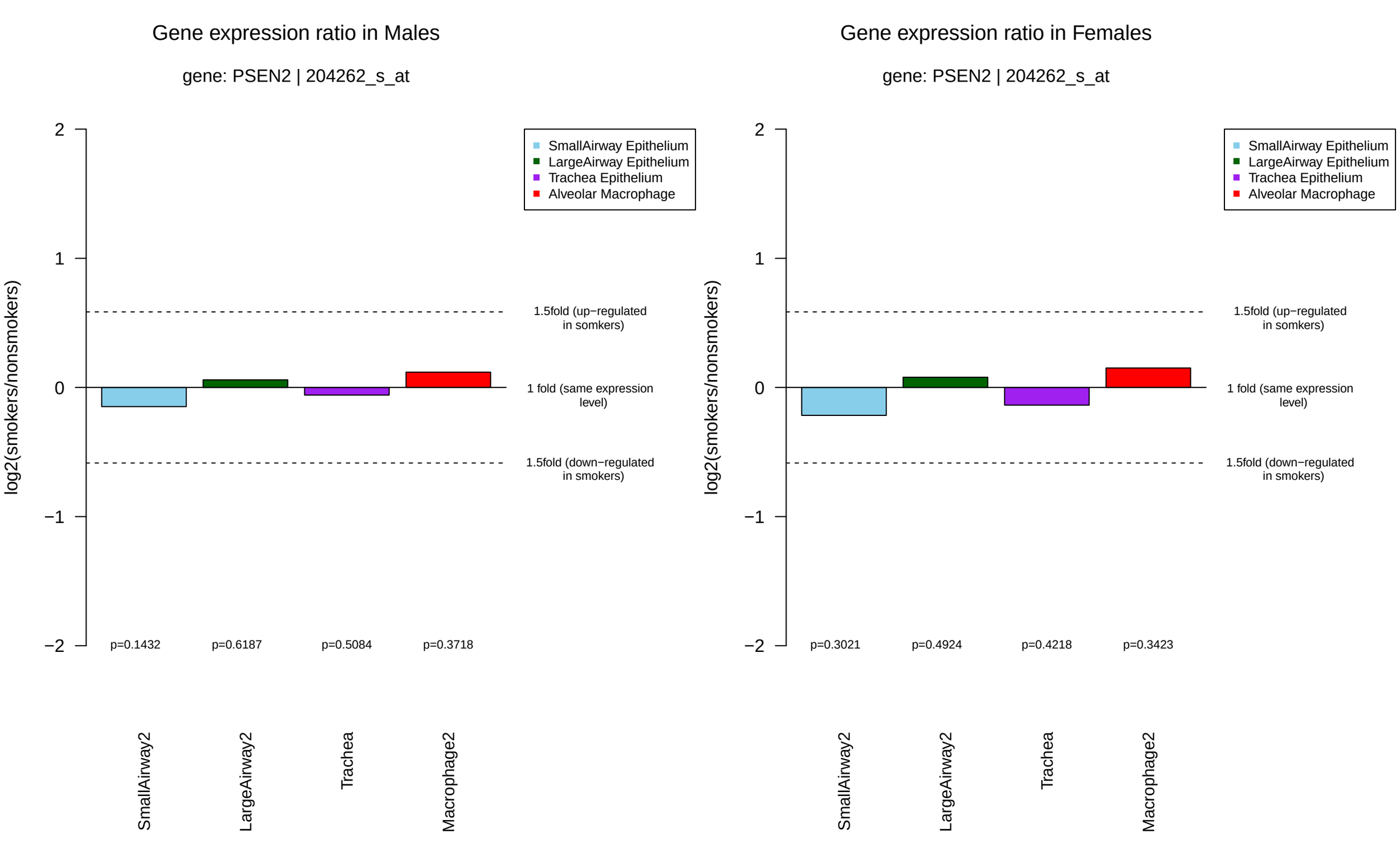
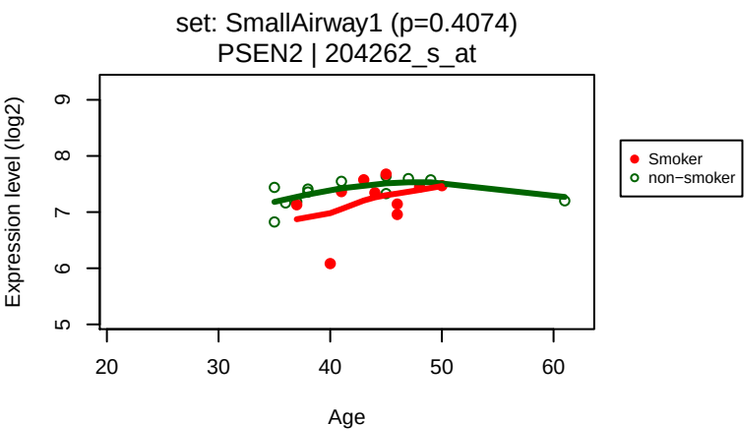
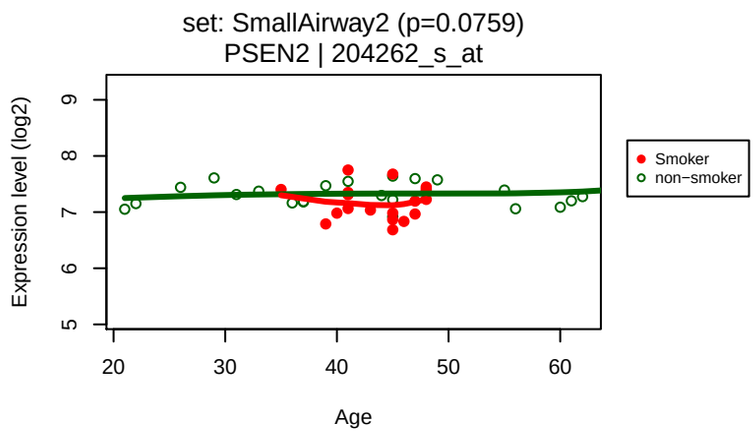
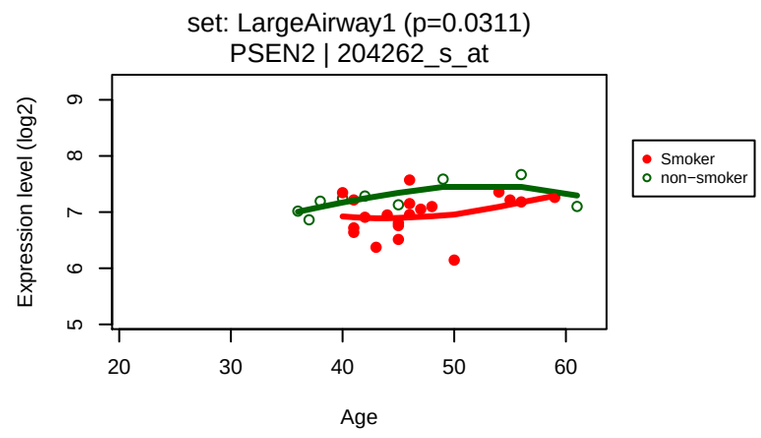

Gene expression of smokers and nonsmokers
gene: PSEN2 | 204262_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4074
p=0.0759
p=0.8626
p=0.0311
p=0.3831
p=0.515
p=0.6725
p=0.1898
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

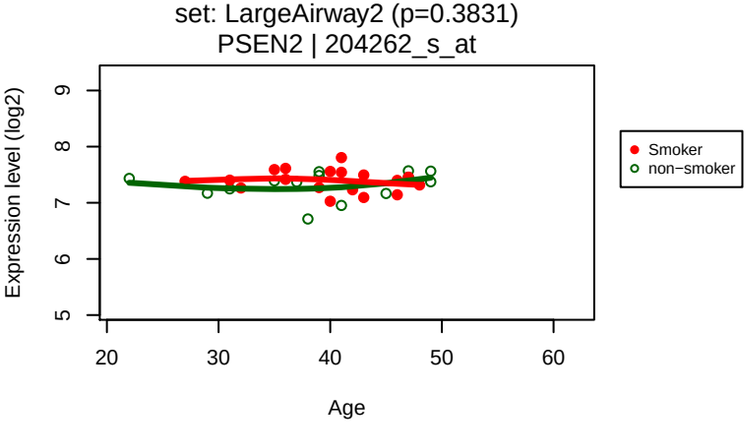

Gene expression ratio (smokers/nonsmokers)
gene: PSEN2 | 204262_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4074
p=0.0759
p=0.8626
p=0.0311
p=0.3831
p=0.515
p=0.6725
p=0.1898
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

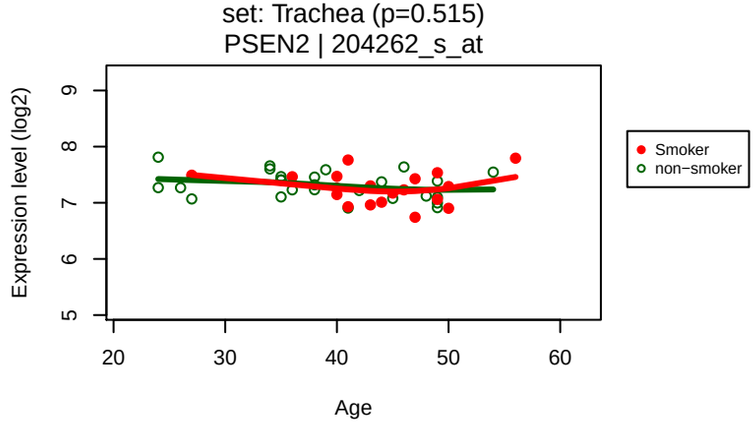

Gene expression ratio in Males
gene: PSEN2 | 204262_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1432
p=0.6187
p=0.5084
p=0.3718
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PSEN2 | 204262_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3021
p=0.4924
p=0.4218
p=0.3423
SmallAirway2
LargeAirway2
Trachea
Macrophage2

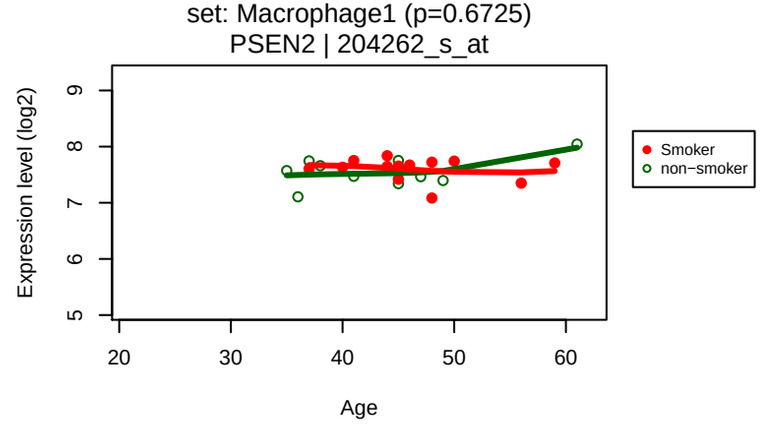

set: SmallAirway1 (p=0.4074)
PSEN2 | 204262_s_at
Expression level (log2)
Age
Smoker
non−smoker

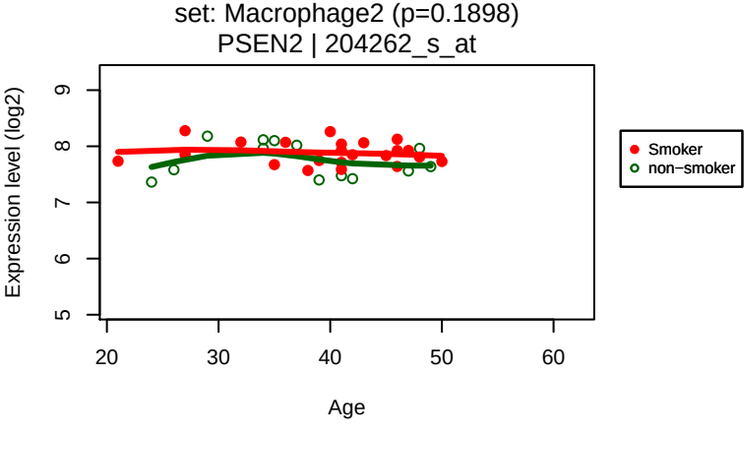

set: SmallAirway2 (p=0.0759)
PSEN2 | 204262_s_at
Expression level (log2)
Age
Smoker
non−smoker

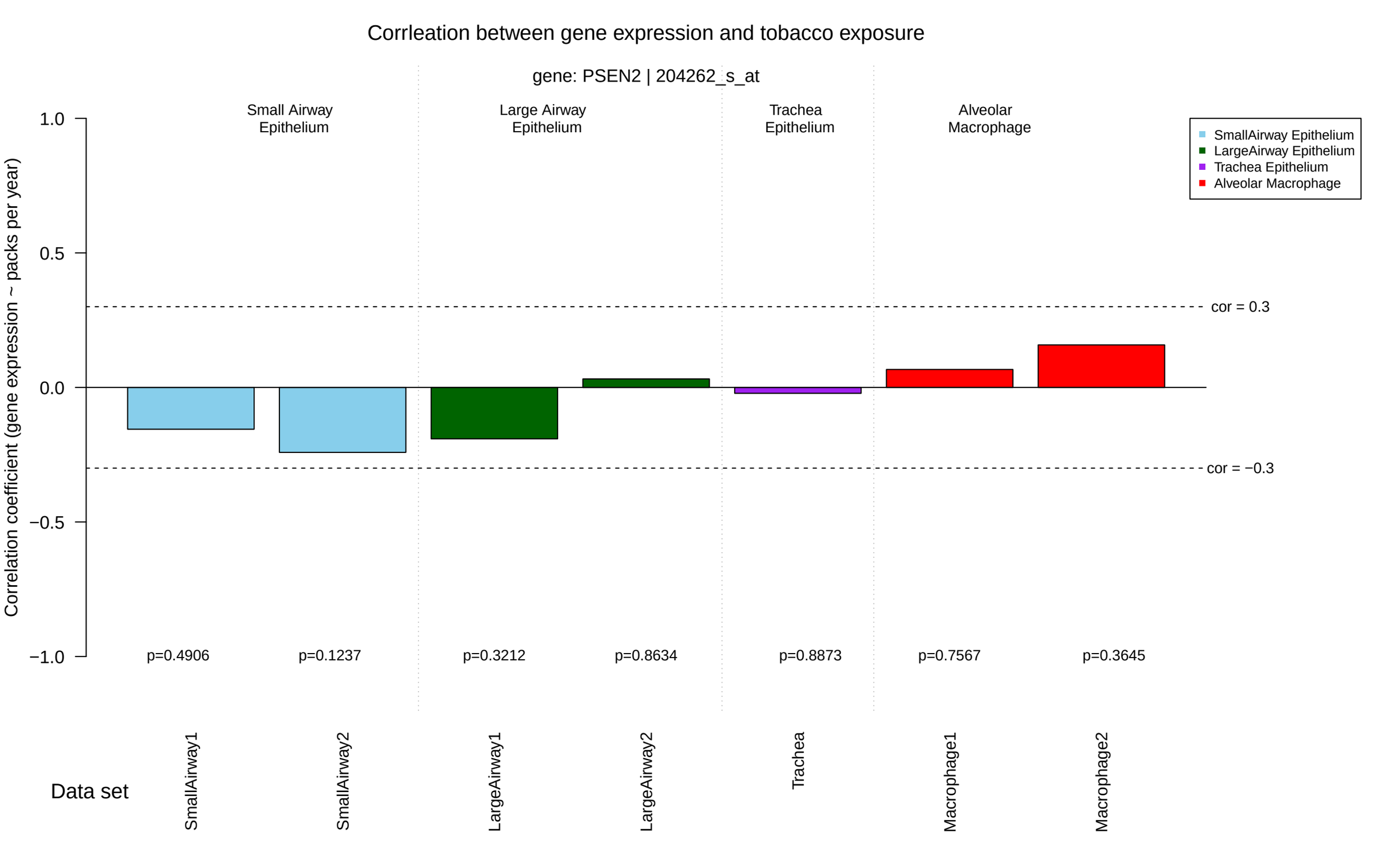

set: LargeAirway1 (p=0.0311)
PSEN2 | 204262_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3831)
PSEN2 | 204262_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.515)
PSEN2 | 204262_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6725)
PSEN2 | 204262_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1898)
PSEN2 | 204262_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PSEN2 | 204262_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4906
p=0.1237
p=0.3212
p=0.8634
p=0.8873
p=0.7567
p=0.3645
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2